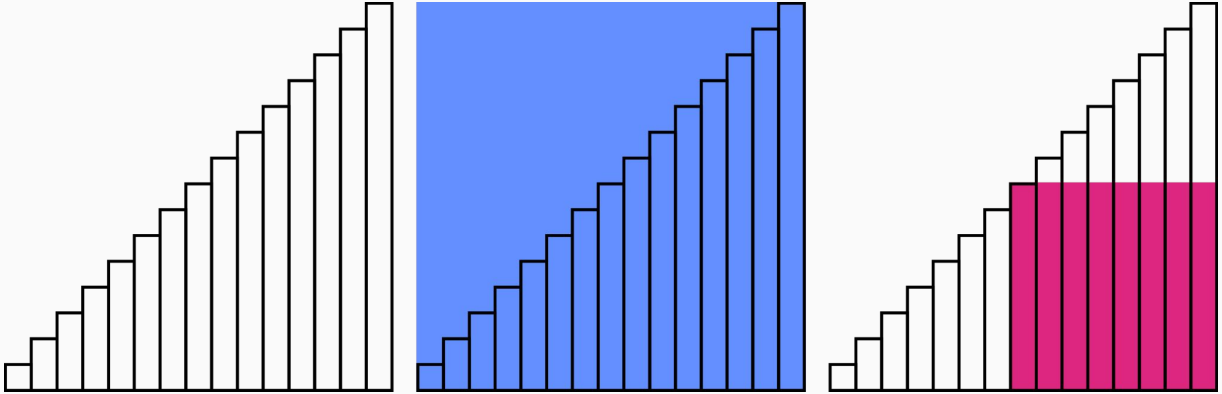
Proof by Induction

Prof. Timothy Sun

A summation formula

We gave upper and lower bounds to prove that

$$1 + 2 + 3 + \ldots + \boldsymbol{n} = \Theta(\boldsymbol{n}^2)$$



A summation formula

But what about an exact formula?

$$1 + 2 + 3 + \ldots + \boldsymbol{n} = \boldsymbol{n}(\boldsymbol{n}+1)/2, \text{ for all } n \geq 1.$$

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Sometimes one can use ***proof by induction*** if there's a recursive structure:

$$(1 + 2 + 3 + \ldots + (\boldsymbol{n}-1)) + \boldsymbol{n}$$

Proof by induction

Every induction proof has two steps:

- Base case (prove for n = starting value)
- Inductive step (***if*** true for $n \leq k$, prove it's true for $n = k+1$, for every k)

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Assuming that it's true for $n \leq k$ is called the **inductive hypothesis**.

Proof by induction

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Base case: check $\boldsymbol{n} = 1$.

- LHS: 1
- RHS: $1(1+1)/2 = 1$

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LHS = RHS

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Proof by induction

Want to prove: 1 + 2 + 3 + … + ***n*** = ***n***(***n***+1)/2, for all ***n*** ≥ 1.

Inductive step: ***if*** formula is true when ***n*** ≤ ***k***, then it's true when ***n*** = ***k***+1

$$\begin{aligned}\sum_{i=1}^{k+1} i &= \left(\sum_{i=1}^{k} i\right) + (k+1) \\ &= \left(\frac{k(k+1)}{2}\right) + (k+1) \\ &= \left(\frac{(k+1)((k+1)+1)}{2}\right)\end{aligned}$$

Why does induction work?

“Isn’t induction circular? You’re assuming [in the inductive hypothesis] the thing you’re trying to prove!”

Emphasis on the word “if”:

“***<u>if</u>*** formula is true when $n \leq k$, then it’s true when $n = k+1$”

Inductive step is conditional

The following statement is true:

> "If $\boldsymbol{n} = \boldsymbol{n}+1$, then $\boldsymbol{n}+1 = \boldsymbol{n}+2$,"

but this is never true when $\boldsymbol{n}$ is an integer.

Why does induction work?

The “domino” analogy:

- Domino ***i*** is the statement for ***n*** = ***i***.
- Each domino knocks over the next one (inductive step).
- The chain reaction starts when knocking over the first domino (base case).

A coding analogy?

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FormulaChecker():
    n := 1
    while True
        if Check(n) = False
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```
Check(n):
    sum := 0
    for i = 1, ..., n
        sum := sum + i
    return sum = n(n + 1)/2
```

A coding analogy?

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```
Check(n):
    if i = 1
        return 1 = 1(1 + 1)/2
    else
        sum := (n − 1)(n)/2
        sum := sum + n
        return sum = n(n + 1)/2
```

We could store “sum,” but we can exploit the previous check passing

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Check(n):
    return True
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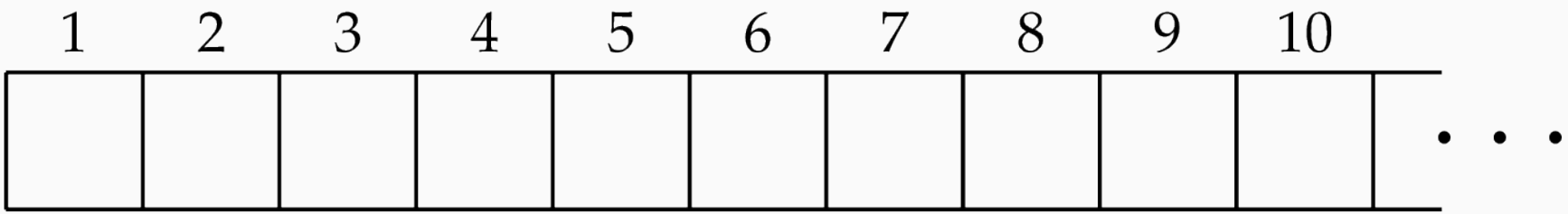
Verifying each branch, we can simplify further

An induction puzzle - weak induction

Will all the boxes be checked under the following rules?

- Box 1 is checked.
- **<u>If</u>** Box ***i*** is checked, then so is Box ***i***+1.



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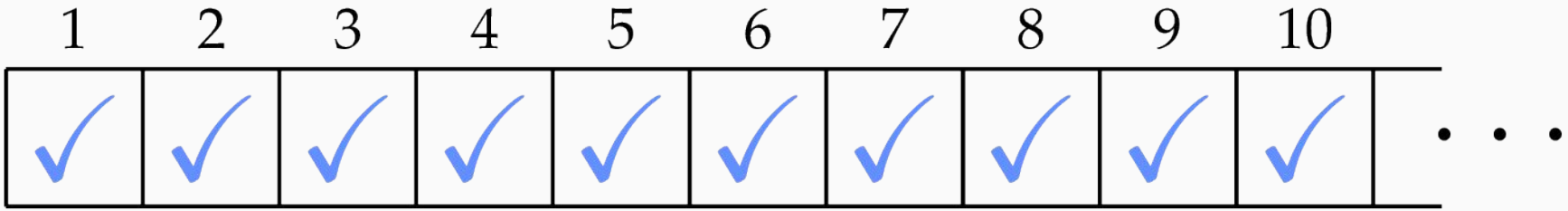

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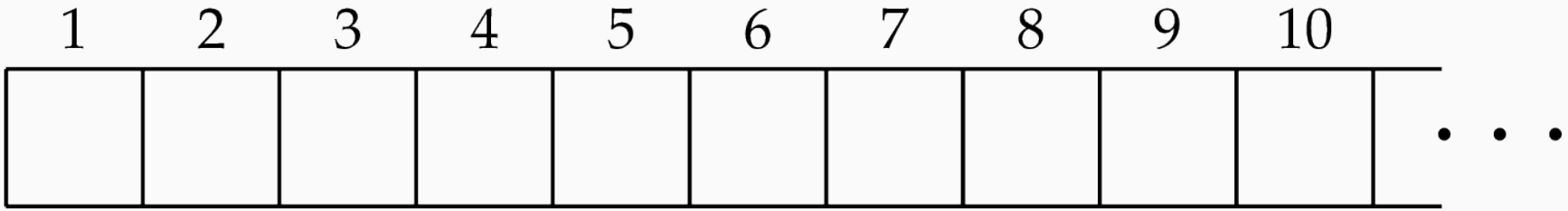

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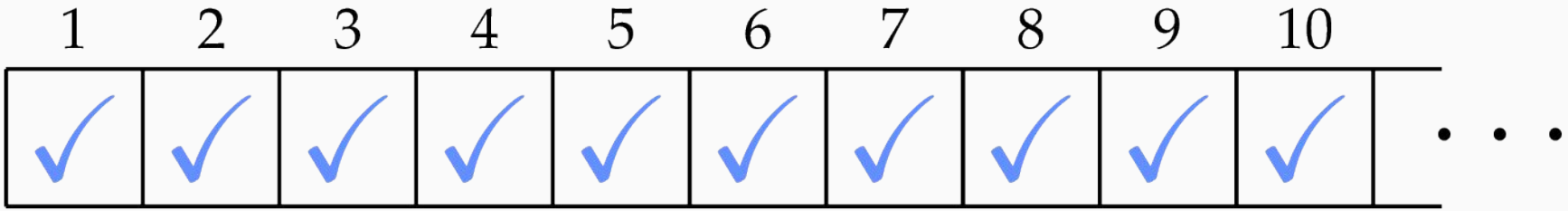

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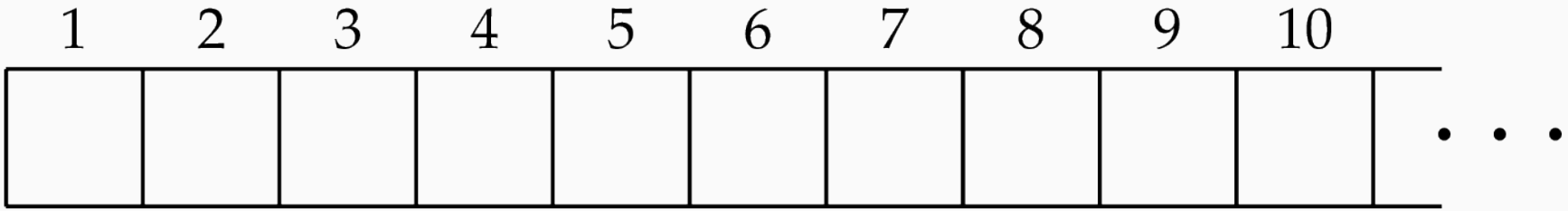

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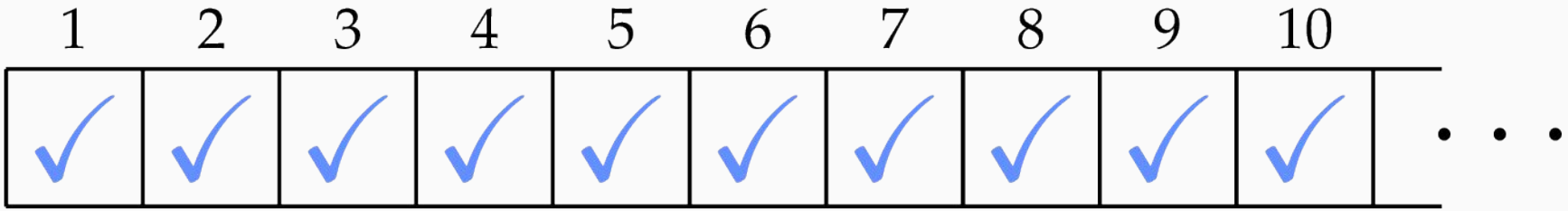


Pitfalls

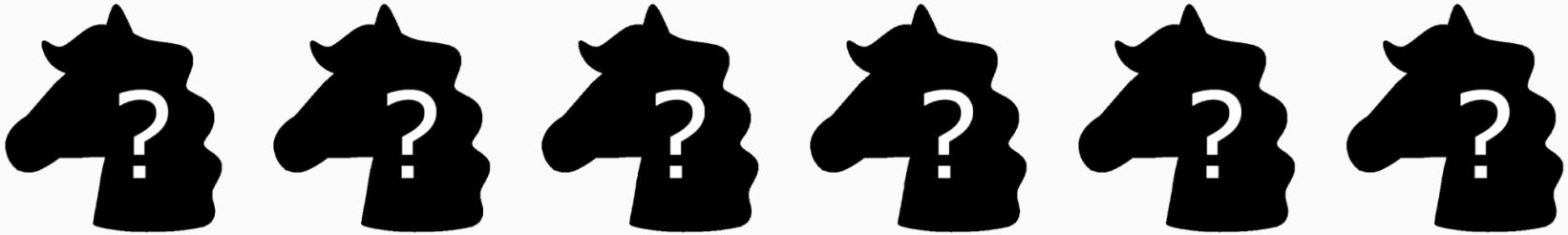
Some fallacious proofs by induction:

- “I’m 2/3 Chinese and 1/3 Latino.”
- “All horses are the same color.”
- “Everyone is essentially bald.”

Monochromatic horses

What's wrong with this inductive step?:

If true for some ***k***, like ***k*** = 5, try proving for sets of ***k*** + 1 = 6 horses.



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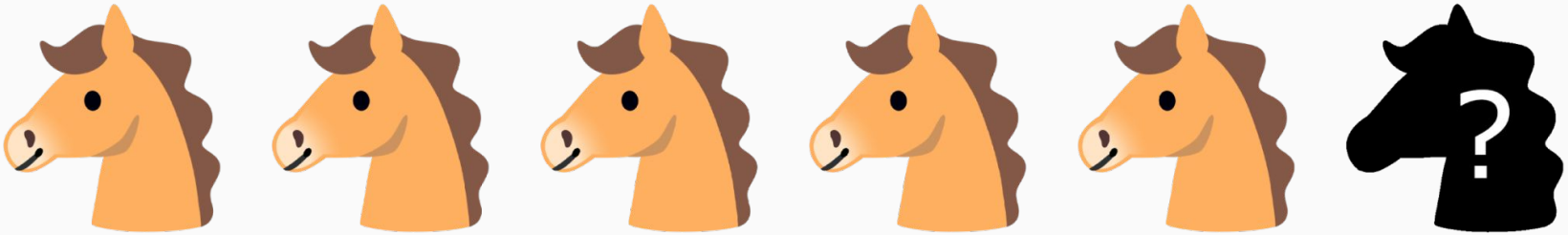

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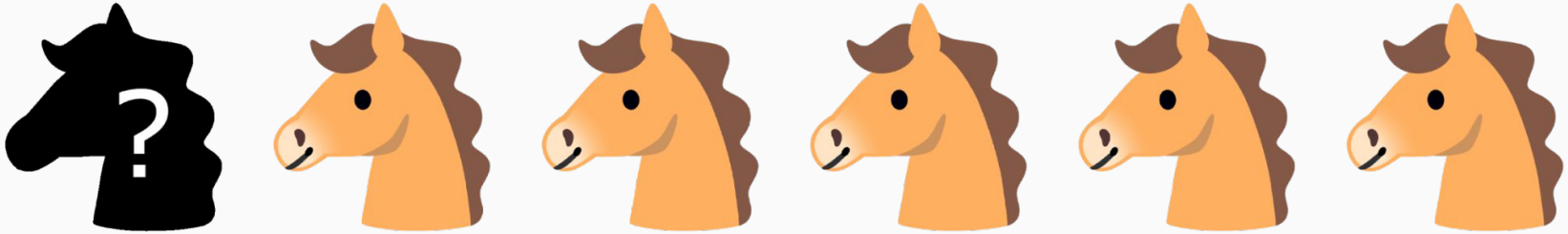

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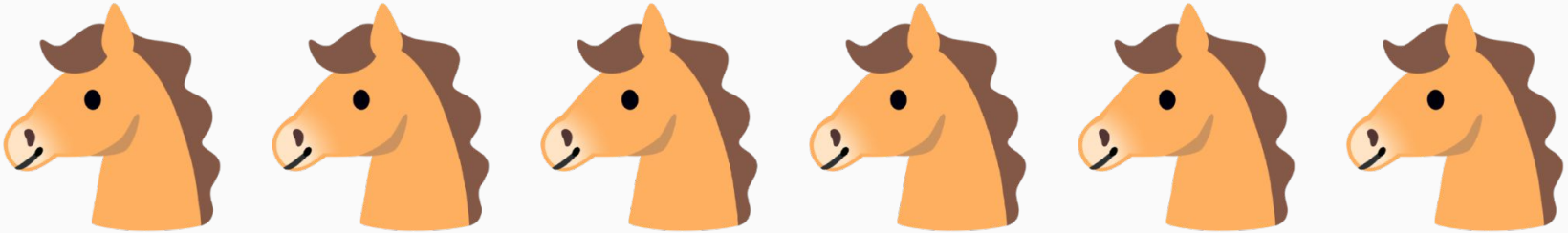

Therefore, all six horses are the same color!

Proof by induction

A few more examples:

- Prove that n^3+2n is divisible by 3 for every positive integer n.
- Find and prove a formula for

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \ldots + \frac{1}{n \cdot (n+1)}$$

- Show that if you have an unlimited collection of 3-cent and 5-cent stamps, you can make any postage ≥ 8 cents.

Revisiting some calculus…

Formally proving $\lim_{x \to \infty} \frac{x^n}{e^x} = 0$ for all ***n*** ≥ 0 using induction.

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Base case: when ***n*** = 0, we can evaluate the limit directly:

$$\lim_{x \to \infty} \frac{x^0}{e^x} = \lim_{x \to \infty} \frac{1}{e^x} = 0$$

Revisiting some calculus…

Inductive step: if it's true for ***n*** ≤ ***k***, use l'Hopital's rule once to show that it's true for ***n*** = ***k*** + 1:

$$\begin{aligned}\lim_{x\to\infty}\frac{x^{k+1}}{e^x} &= \lim_{x\to\infty}\frac{(k+1)x^k}{e^x}\\ &= (k+1)\left(\lim_{x\to\infty}\frac{x^k}{e^x}\right)\\ &= (k+1)\cdot 0\\ &= 0\end{aligned}$$

Well-ordering principle

“For every ***non-empty*** set of positive integers, there is a smallest element.”

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Equivalent to induction:

- "If it were false, there would be a smallest counterexample…"